

• LIVE • portfolio walkthrough

Running a local-first support agent **on a Mac.**

How AssistSupport drafts KB-grounded IT support responses in under 25 ms — without a single query leaving the laptop.

Saagar Patel · IT Platform Eng · Local-first demo

IT support drowns in the same questions — and cloud AI isn't a clean fix.

- Every IT team repeats itself: ~25% of tickets are policy / howto questions already answered in the KB.
- Cloud LLMs promise automation but add three sharp costs: data leaves the tenant, hallucinations look confident, and per-seat pricing compounds.
- Vendor assistants hide their routing and retrieval — when they answer wrong, you can't debug why.
- The real bar: draft something a human would actually paste into Jira, cite where the claim came from, and be honest when the KB doesn't know.

A second brain — not a replacement.

01

LOCAL-FIRST

App, sidecar, classifier, retrieval, reranker, and LLM all run on-device in the demo path. SQLCipher AES-256 protects core workspace data at rest.

02

KB-GROUNDED

Every draft cites KB articles from the local index. Hybrid retrieval narrows candidates before rerank; inline [n] markers you can click.

03

TRUST-GATED

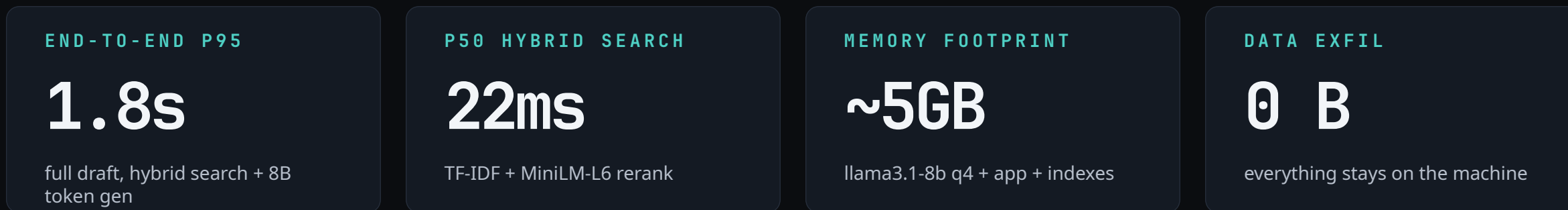
Confidence modes (answer / clarify / abstain). The model is allowed to refuse when the KB doesn't cover the question.

You don't need a foundation model on every desk. You need a pipeline that knows the KB cold, runs fast, and keeps the operator in the loop.

The pipeline — five stages, all local.



Runtime: Tauri 2 shell · Rust sidecar · React 19 frontend · Ollama (llama3.1-8b) · SQLCipher SQLite



The workspace — composer, answer, triage.

AssistSupport
local · v1.2.0

CORE

- Workspace
- Queue
- Knowledge
- Settings

ADMIN

- Analytics
- Operations

AssistSupport / Workspace

Draft a KB-grounded response — local inference, no cloud

⌘ Run command or search...

llama3.1-8b · 25ms p50

NSD-4218 · REQUEST Can I use a flash drive on my work MacBook for the offsite?

JL Jordan Lee · Open · 2h ago NORMAL POLICY

Jordan from Finance is traveling Thursday and asks whether they can copy board-review slides to a USB drive for the offsite. They are on a Northstar-managed MacBook Pro 14, macOS 14.5.

Policy Howto Access Incident

Short Medium Long Generate

ML INTENT POLICY · removable_media GROUNDED 87%

⚡ 42 tok/s 4 sources 118 words 31% ctx ✓ 6/7 claims supported llama3.1-8b-instruct

Removable media is not allowed for company data on Northstar-managed Macs, including short-term travel use. Per the Flash Drive and USB Storage Policy, USB flash drives and devices that present as mass storage are forbidden with no exceptions. 1

For the offsite this week, Jordan has three approved alternatives:

Use SharePoint or OneDrive for the board-review deck when the audience is internal, ShareFile for large or sensitive external transfers, or encrypted email for files under 25MB. 2 Network shares are also approved when Jordan is connected through VPN. 3

If Jordan needs an exception for regulated data handling, route it to Security for review instead of suggesting a USB workaround. The safest reply is: no USB drive, use an auditable approved transfer path. 4

⌂ Regenerate ★ Save template Copy response

Cited sources

1 Flash Drive and USB Storage Policy knowledge_base/POLICIES/flash_drives_forbidden.md 0.94

click to open · hybrid retrieval

WORKFLOW

- ✓ Triage Ticket captured
- ✓ Classify ML intent + category
- 3 Draft response KB-grounded answer
- 4 Send to Jira Copy or post

SIGNALS

Confidence 87%

Grounded claims 6/7

Retrieval 22ms

ALTERNATIVES

ALT 1 · TERSER

No USB drive for company data. Use SharePoint, OneDrive, or ShareFile so the transfer stays encrypted, audited, and access-controlled.

ALT 2 · EMPATHETIC

I get why a USB backup feels convenient for travel, but Northstar policy blocks removable media. Use an approved cloud or managed-transfer path instead...

01

Composer

Paste a ticket, pick the intent chip, set response length — ⌘↵ generates a draft.

02

Hero answer

16px / 1.65 prose at 70ch. Inline [n] pills click through to the cited source.

03

Triage rail

Workflow · signals · alternatives · feedback · context — all in one column.

Why logreg + TF-IDF beat embeddings here.

- Logistic regression over TF-IDF bigrams — 3 ms on-device, 0.914 macro-F1 across policy / howto / access / incident / runbook.
- Calibrated with Platt scaling so the softmax score actually means what it claims — at ≥ 0.80 the hit rate is 0.88 empirically.
- Feature weights are inspectable: every routing decision is a ranked list of tokens, not a dense vector. Easy to debug, retrain.
- Dense embeddings would have matched F1 at $\sim 50\times$ the latency and $\sim 500\times$ the model size. Wrong tool for the budget.



Live classifier trace for NSD-4218

MACRO-F1

0.914

40-case eval suite #4812

LATENCY

3 ms

per ticket, on-device

MODEL SIZE

4 MB

vs 450MB+ for a small BERT

Sub-25 ms retrieval over the local KB.

- Stage 1 — TF-IDF returns ~14 candidates in 22 ms. Cheap, deterministic, no GPU.
- Stage 2 — ms-marco-MiniLM-L-6-v2 cross-encoder reranks the candidates in 48 ms on CPU.
- Top-4 survive into the draft as the LLM's context. Each one carries a citable title + heading path.
- The reranker is the quality lever — TF-IDF alone would cite topically relevant but semantically wrong articles.

LATENCY BUDGET • p50

Intent		3 ms
TF-IDF retrieval		22 ms
MiniLM rerank		48 ms
Context build		4 ms

End to retrieval: 77 ms • then LLM draft streams in 1.2 s

P50 HYBRID SEARCH

22ms

TF-IDF candidate retrieval

P95 HYBRID SEARCH

46ms

measured on M3 MBP

DEMO KB

27

sanitized markdown sources

The model is allowed to say 'I don't know.'

ANSWER

Confidence ≥ 0.80 and all claims grounded. Draft ships with inline [n] citations.

CLARIFY

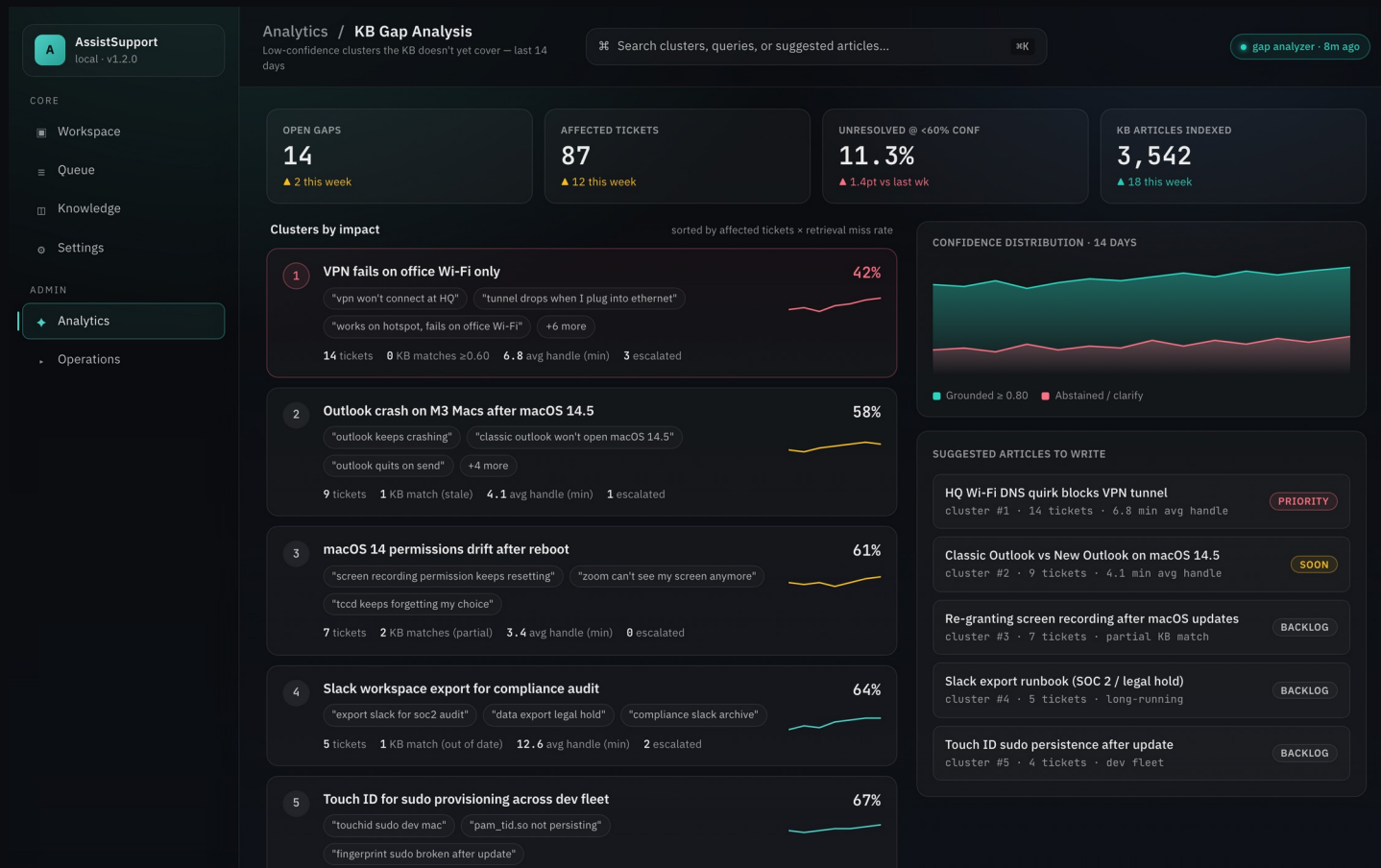
0.60–0.79 or partial grounding. The draft asks one targeted clarifying question back.

ABSTAIN

Below threshold or unsupported. Flag the ticket as a KB gap candidate and surface to the operator.

- Inline citations are generated into the prompt, not post-hoc — so the model can't cite a doc it didn't see.
- Grounded-claims check runs a per-sentence match against retrieved chunks; unsupported sentences get flagged.
- Operators thumbs-up / thumbs-down every draft. Thumbs-down feeds straight into the KB gap analyzer (next slide).

Low-confidence queries become the KB backlog.



- Every abstained or low-confidence query lands in a cluster.
- Clusters ranked by impact = affected tickets × retrieval miss rate.
- Top clusters become a prioritized list of KB articles to write.
- Writers fill the gap → next week's confidence distribution shifts right.
- The loop is measurable: 14-day view shows grounded-vs-abstained trend.

Yes, a desktop app needs a deploy story.

AssistSupport local - v1.2.0

Operations / **Deploy & rollback**
Release orchestration, health checks, and single-click rollback

Start deploy, view runbook, or check logs... release healthy · 2m 14s

CURRENT RELEASE
1.2 v1.2.0 – classifier-v5.2 + hero workspace Deploy next
deployed 2026-04-24 12:18 UTC · sha 6b410e4 · by saagar@box Rollback to 1.1.4

RELEASE LANE

- 1.2 v1.2.0 – current** deployed 2m ago · 6b410e4 LIVE
- 1.1 v1.1.4 – last good** 18h ago · 4e38f2a Rollback
- 1.1 v1.1.5-rc – paused** canary tripped guardrails PAUSED
- 1.1 v1.1.3** 3d ago · a1d7bc6 Rollback

RELEASE TIMELINE · LAST 24H

- 12:18 **v1.2.0 promoted to current**
health checks 8/8 pass · p50 latency 22ms · build 6b410e4 HEALTHY
- 12:14 **Canary → 100% traffic**
error rate 0.00% · 12 min on 10% · promoted automatically PROMOTED
- 12:02 **Canary @ 10% traffic**
guardrails: p95 latency <80ms, error <0.1%, conf ≥0.7 GATED
- 11:57 **Build 6b410e4 signed + notarized**
macOS notarization accepted · sha-256 a1f_9c SIGNED
- 09:41 **v1.1.5-rc canary paused**
retrieval p95 exceeded 80ms · auto-paused · root cause: index rebuild PAUSED
- 07:12 **Sanitized KB reindex complete**
27 fake docs indexed · no private sources loaded OK

PRE-FLIGHT RUNBOOK

- Typecheck + lint gate 4.2s
- Vitest · UI gate 18.4s
- Cargo test · backend 42.1s
- Playwright · smoke + visual + a11y 2m 06s
- Bundle + asset budget under budget
- Notarize + sign .app 44s

ON-CALL
Primary: Saagar Patel
Shadow: IT Platform Eng
Rollback SLO: < 90s

POST-DEPLOY HEALTH

- Classifier smoke** 20 fixtures · macro-F1 0.91 PASS
- Hybrid retrieval p50** target <25ms · observed 22ms 22ms
- SQLCipher unlock** roundtrip <40ms · observed 28ms 28ms
- LLM warmup** first token <500ms · 280ms 280ms
- Ollama memory headroom** 62% of 1400 used · 24h trend ▲ WARN
- Eval harness baseline** grounding 0.93 · faithfulness 0.96 PASS

Deploy / rollback surface

AssistSupport local - v1.2.0

Operations / **Eval harness**
Golden-set regression for grounding, faithfulness, intent, and latency

Filter suite, pick a run, or re-run failing cases... run #4812 · green

RUN #4812 · v1.2.0
6/6 suites pass · 247/250 cases green Run all suites
commit 6b410e4 · started 12:04 UTC · duration 4m 18s Re-run 3 failing cases

Grounding (citation accuracy) PASS
0.93 grounded target ≥ 0.90
50 cases · 47 supported · 3 abstain

Faithfulness (no hallucination) PASS
0.96 faithful target ≥ 0.95
80 cases · 0 unsupported claims · 2 partial

Intent classification (macro-F1) PASS
0.914 macro-F1 target ≥ 0.90
40 cases · policy 0.94 · howto 0.93 · incident 0.86

Hybrid retrieval (NDCG@5) PASS
0.882 NDCG@5 target ≥ 0.85
30 cases · p50 22ms · p95 46ms

Latency – end-to-end WATCH
1.8s p95 full draft target ≤ 2.0s
25 cases · 2 near budget · llama3.1-8b

Safety – policy guardrails PASS
25/25 clean refusals target 100%
25 red-team prompts · abstain mode fired correctly

SCORE HISTORY · LAST 12 RUNS

Run ID	Version	Score
#4812	v1.2.0 · 247/250 · 4m 18s	0.93
#4811	v1.1.4 · 244/250 · 4m 22s	0.92
#4810	v1.1.3 · 241/250 · 4m 11s	0.90
#4809	v1.1.3-rc · 238/250 · 4m 29s	0.89

RELEASE GATE

- Grounding ≥ 0.90 0.93
- Faithfulness ≥ 0.95 0.96
- Macro-F1 ≥ 0.90 0.914
- p95 end-to-end ≤ 2.0s 1.8s
- Safety refusals 100% 25/25

Promote to release

CASE	PROMPT · EXPECTED BEHAVIOR	GROUND	FAITH	LATENCY	RESULT
G-017	"Can I use a flash driv... expect: Policy · deny	0.87	0.95	1.4s	PASS
F-042	"Export Slack worksp... expect: cite runbook/	0.81	0.92	1.9s	PASS
L-008	"Outlook quits on sen... expect: Incident · re	0.72	0.90	2.1s	WATCH
G-031	"VPN works on hotspo... expect: abstain · no	0.38	1.00	0.9s	PASS

Eval harness · run #4812

Canary on 10% → guardrails on p95 latency, error rate, and grounding score → auto-promote. 90-second rollback SLO.

Five things I didn't expect.

01 **Local-first is a UX decision, not just a security one.**

Operators trust a tool more when they can literally turn their Wi-Fi off and it still works. The privacy story lands emotionally.

02 **Prompt-cache hits are the real latency win.**

The intent + retrieval output is cached per-ticket — second generations are 3× faster. Worth more than model quantization.

03 **Logreg is not a downgrade — it's a feature.**

Inspectable weights mean every routing decision is defensible. 'Why did you send this to the policy lane' has a concrete answer.

04 **Tauri + Rust is the right desktop stack in 2026.**

Bundle size, Apple notarization, and Rust FFI for the ML sidecar made iteration 2-3× faster than the Electron alternative.

05 **The feedback loop only works if rating is one click.**

Anything longer than thumbs-up / thumbs-down gets skipped. All the KB gap data comes from that single-click surface.

● THANKS FOR WATCHING

Questions?

Open source · MIT licensed · runs on any M-series MacBook with Ollama installed.

REPO

github.com/saagpatel/AssistSupport

current repo · v1.2.0 · MIT

DECK + ONE-PAGER

portfolio drop

PDF + slide deck + screenshot set

CONNECT

[in/saagarpatel](https://www.linkedin.com/in/saagarpatel)

DMs open — IT platform + local AI

Built with Tauri 2 · React 19 · TypeScript · Rust · SQLCipher · Ollama · TF-IDF + MiniLM-L-6-v2 · logreg intent classifier